

# Monitor for 6V/12V Batteries

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Rechargeable batteries of 6V and 12V are used in a large number of applications. It is imperative that these are maintained properly to get maximum life out of them. Further, their permissible number of charge-discharge cycles must be fully utilised. Here is a circuit that gives a visual as well as an audible alarm if the battery voltage is higher or lower than acceptable limits, so that corrective action can be taken.

## Circuit and working

As shown in Fig. 1, the circuit is built around dual-operational amplifier LM358 (IC1), hex inverting Schmitt trigger 74HC14 (IC2) and a few other components. The circuit can be divided into two parts—the input unit built around two operational amplifiers (op amps) in IC1 working as comparators, and audio alarm unit built around two RC oscillators and a transistor.

Voltage from the battery under test (BUT) is applied to connector CON1. It is divided by five by R1 and R2 for comparison with the threshold levels. The power supply of the circuit is limited to 5V  $\pm$ 2%. SPDT

switches S1 and S2 simultaneously select between the preset threshold for 6V and 12V rechargeable batteries. Potentiometers VR1 and VR2 set the highest allowable input voltage. For example, for a 12V battery, if you wish to activate the visual and audio alarms at 14.5V, you should set the potentiometer VR2 reference voltage to  $14.5V/5 = 2.9V$ .

Potentiometers VR3 and VR4 set the lowest allowable input voltage. For example, for a 12V battery, if you wish to activate the visual and audio alarms at 10.8V, you should set the potentiometer VR4 reference voltage to  $10.8V/5 = 2.16V$ . Similarly, you can obtain the reference voltages for a 6V battery by setting potentiometers VR1 and VR3.

When the input voltage is above the threshold set by potentiometers VR1 or VR2, the output at pin OUT1



of IC1 becomes low and the LED2 indicating a high voltage switches on. The output of gate N1 of IC2 becomes high and the RC oscillator built around gate N2 oscillates. The output, which is a square wave with frequency of approximately 1kHz, is

## PARTS LIST

### Semiconductors:

- IC1 - LM358 dual op-amp
- IC2 - 74HC14 hex Schmitt trigger inverter
- T1 - BC547 npn transistor
- D1-D4 - 1N4148 signal diode
- LED1-LED3 - 5mm LED

### Resistors (all 1/4-watt, $\pm$ 5% carbon):

- R1 - 12-kilo-ohm
- R2 - 3-kilo-ohm
- R3 - 2-kilo-ohm
- R4, R11, R13 - 470-ohm
- R5, R6 - 1-mega-ohm
- R7, R8 - 1-kilo-ohm
- R9, R10 - 2-kilo-ohm
- R12 - 100-ohm
- R14 - 22-kilo-ohm
- VR1-VR4 - 10-kilo-ohm potentiometer

### Capacitors:

- C1 - 470 $\mu$ F, 16V electrolytic
- C2, C3 - 1 $\mu$ F, 16V electrolytic
- C4, C5 - 0.33 $\mu$ F ceramic disk

### Miscellaneous:

- CON1, CON2 - 2-pin terminal connector
- S1, S2 - SPDT switch
- LS1 - 32-ohm headphone speaker
- 6V/12V battery under test
- 5V  $\pm$ 2% power supply

## Test Points

| Test point | Details                     |
|------------|-----------------------------|
| TP0        | 0V                          |
| TP1        | 5V                          |
| TP2        | Around 1kHz when LED2 glows |
| TP3        | Around 1kHz when LED3 glows |

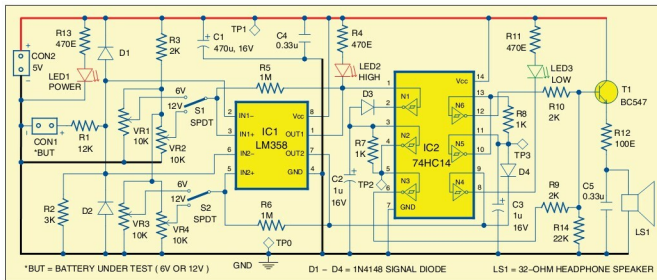


Fig. 1: Circuit diagram of monitor for 6V/12V batteries